Assignment - II

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Course Name: Deep learning

1) Explain in detail about working of Alexnet and Resnet architecture with suitable diagram.

1) Alexnet architecture

Alexnet is one of the first deep convolutional neural networks that made deep learning Popular by winning the Imagenet 2012 competition

- input size : 227 X 227 X 3 (RGB image)
- layers : 8 layers (5 conv + 3 fully connected)
- users : Relu, Dropout, max pooling

Architecture layers

1. input layers

image is resized to 227 X 227 X 3

Example: A cat image.

2. convolutional layer

filters : 96 filters (11 X 11)

stride : 4

output : feature maps

Detects basic features like edges, colors.

3. Activation (Relu)

Applies Relu = max (0, x)

makes training faster

4. Max Pooling

Reduces size of feature map

keeps important features.

5. convolution Layer 2

Filters : 256 (5×5)

learns more complex patterns

6. convolution layers 3, 4, 5

Filters : 384, 384, 256

Detect higher-level features like shapes, objects.

working flow

input : image of a dog

1. conv1 :- detects edges
2. conv2 :- detects textures
3. conv3 :- detects dog face
4. fc layers → classify
5. output → "Dog"

key features

→ Introduced Relu activation.

→ used apu training.

2) ResNet Architecture

Resnet solved the vanishing gradient problem and allowed very deep networks (50, 101, 152 layers).

core concepts: Residual Block

Instead of learning : $H(x)$

Resnet learns : $F(x) = H(x) - x$

So o/p becomes : $H(x) = F(x) + x$

Architecture

initial layers

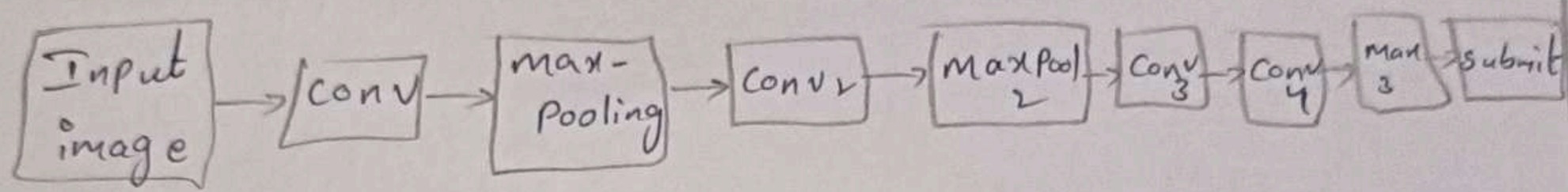
conv layers + max pooling

Residual Blocks

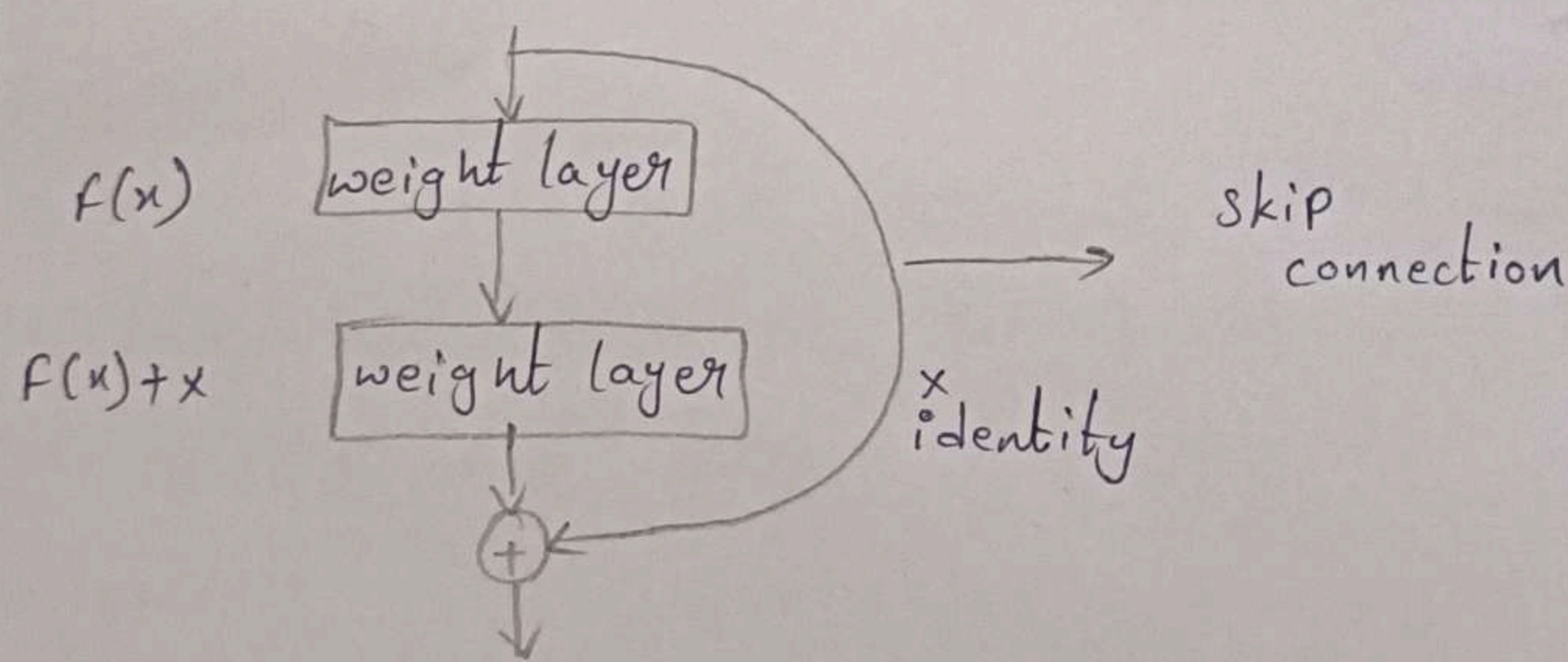
many stacked residual blocks

Each block.

Alex net:



Resnet:-



2) Elabrate GRU (gated recurrent unit) and long short term memory (LSTM) architecture in detail.

1) why LSTM and GPU are needed

Traditional RNNs have a major problem vanishing Gradient problem.

- They forgot long-term information
- cannot learn dependencies over long sequences

Ex:-

"I grew up in France ... : speak Fluen_"

RNN may forget "France" → wrong prediction LSTM / GRU remember → correct : French

2) LSTM

→ LSTM is a special type of RNN designed to remember long-term dependencies using a memory cell and gates.

components

1) forget gate (f_t)

→ Decides what to forget from previous memory

formula:-

$$f_t = \sigma(w_f(h_{t-1}, x_t) + b_i)$$

0 → forget completely

1 → keep fully.

2 Input Gate (i_t)

Purpose :- Decides what need information to store

$$i_t = \sigma(w_i(h_{t-1}, x_t) + b_i)$$

3. candidate memory ($\tilde{c}_t$)

creates new candidate values

$$\tilde{c}_t = \tanh(w_c(h_{t-1}, x_t) + b_c)$$

working flow

sentence

"The movie was not good"

step-by-step

"movie" → stored

"not" → important

"good" → Positive word

GRU :-

→ GRU is a simplified version of LSTM with fewer gates and faster computation.

core components

1) Purpose controls how much past information to keep

$$z_t = \sigma(w_z(h_{t-1}, x_t))$$

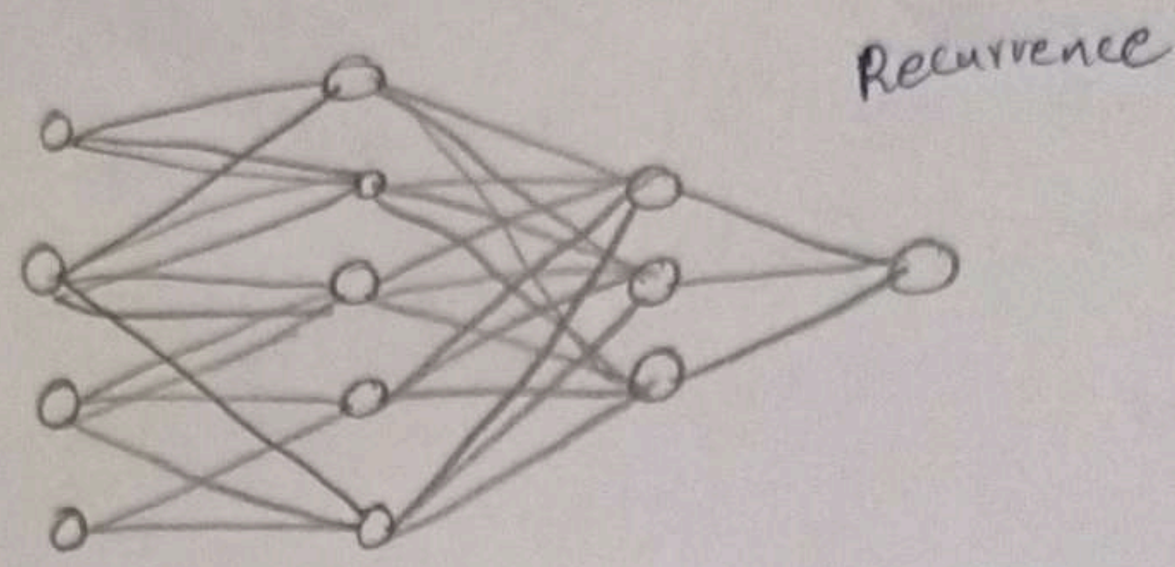
2) Reset Gate

$$z_t = \sigma(w_r(h_{t-1}, x_t))$$

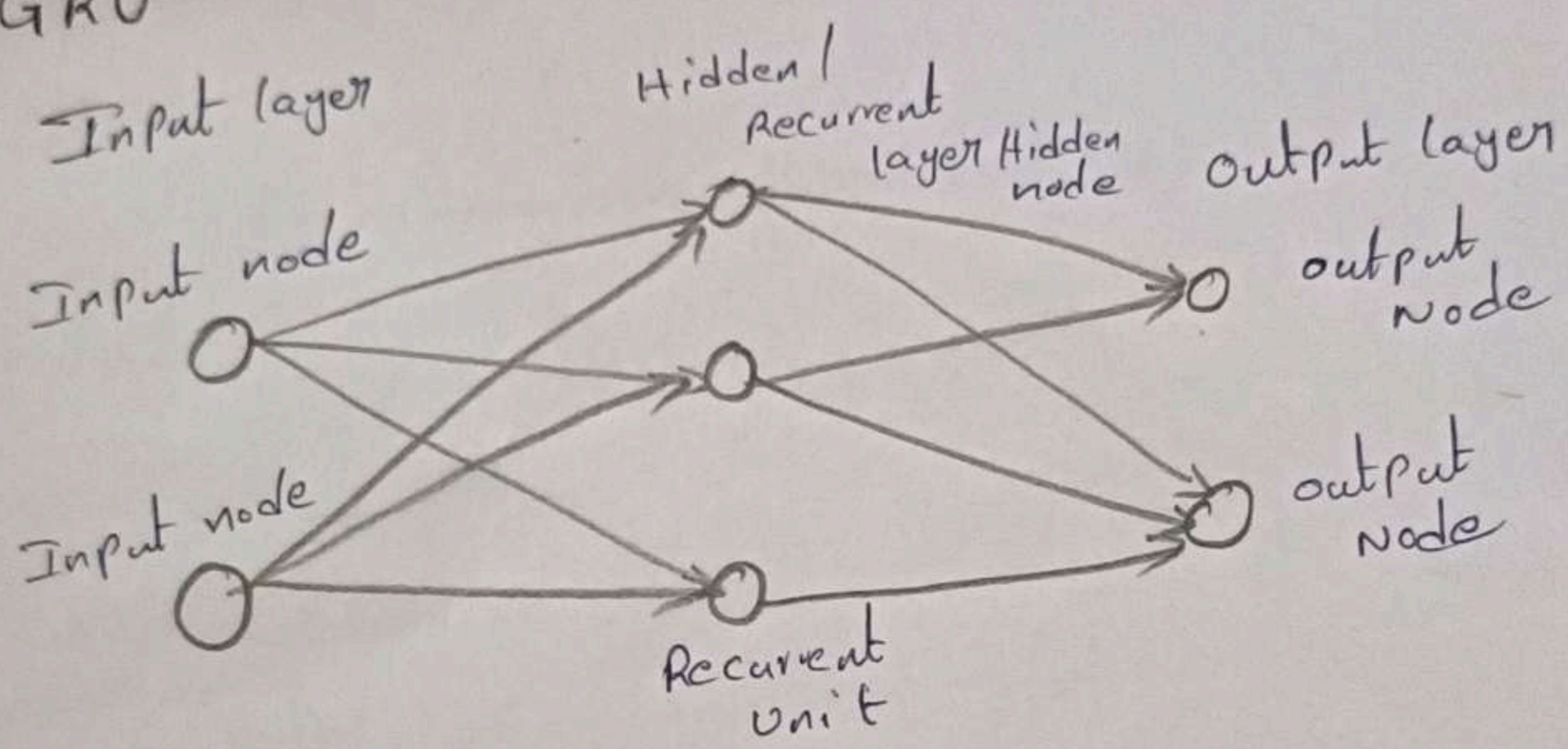
3) candidate Hidden state ($\tilde{h}_t$)

$$\tilde{h}_t = \tanh(w(r_t * h_{t-1}, x_t))$$

LSTM


Recurrence

GRU


Input layer
Hidden / Recurrent layer
Hidden node
Output layer
Input node
output Node
Input node
output Node
Recurrent Unit

Demonstrate spatial Transformer network.

→ spatial Transformer network is a module that allows a neural network to learn spatial transformations like rotation, scaling, translation) of input data automatically.

components:

1. Localization network

<u>Purpose</u>

Learns transformation Parameters (θ)

output:

$$\theta = \begin{pmatrix} \theta_1 & \theta_2 & \theta_3 \\ \theta_4 & \theta_5 & \theta_6 \end{pmatrix}$$

2. Grid generator

creates a sampling grid

- (x^t, y^t): target coordinates
- (x^s, y^s): source coordinates

$(x_i^s, y_i^s) = T_\theta (x_i^t, y_i^t)$

3) sampler

Purpose: Generates transformed image

uses bilinear interpolation

Picks pixel values from input.

Produces final transformed output.

step-by-step working

Ex:- Digit recognition (MNIST)

input image: Rotated digit "6"

step 1:- Input image

Rotated / shifted digit

2) Localization network:

predicts transformation parameters (θ)

Ex:- rotate - 30; center image.

3) Grid generator

creates grids to align digit properly.

4) sampler

Applies transformation

output:- straight, centered digit.

5) CNN classifier

Now easier to classify

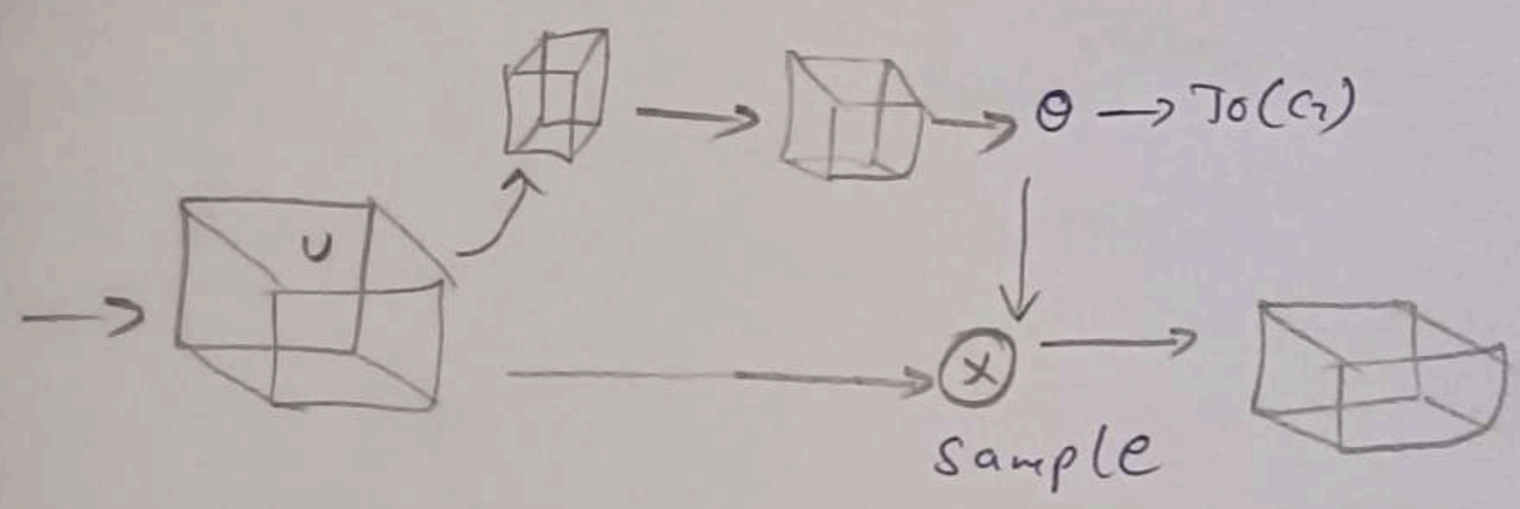
o/p: "6"

mathematical summary

Affine transformation:

$$\begin{bmatrix} x_s \\ y_s \end{bmatrix} = \begin{bmatrix} \theta_1 & \theta_2 & \theta_3 \\ \theta_4 & \theta_5 & \theta_6 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix}$$

Applications

- image classification
- object detection
- face alignment
- medical image analysis

spatial Transformer network.